Business Intelligence Practicals 1–8

Exam-ready Python code with CSV upload options

Practical	Dataset / Source	Upload
1	Sample - Superstore.csv (Power BI)	Power BI
2	Sample - Superstore.csv (Power BI)	Power BI
3	SuperMarket_Analysis_with_Payment_Method.csv	Yes
4	iris.csv	Yes
5	Mall Customers — GitHub	Optional
6	store_sales_regression.csv	Yes
7	customers_ml_logistic_regression.csv	Yes
8	iris.csv	Yes

Practical 1 — Power BI 1

No Python code.
Import Sample - Superstore.csv directly into Power BI.

Practical 2 — Power BI 2

No Python code.
Use Sample - Superstore.csv / Sales data in Power BI.

Practical 3 — Data Analysis

```
import pandas as pd
import matplotlib.pyplot as plt

from google.colab import files
uploaded = files.upload()

df = pd.read_csv("SuperMarket_Analysis_with_Payment_Method.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nStatistical Summary:")
print(df.describe())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nData Types:")
print(df.dtypes)

numeric_columns = df.select_dtypes(include="number").columns

print("\nAverage of Numerical Columns:")
print(df[numeric_columns].mean())

print("\nCorrelation Matrix:")
print(df[numeric_columns].corr())

df[numeric_columns].hist(figsize=(12, 8), bins=15)
plt.suptitle("Distribution of Numerical Variables")
plt.tight_layout()
plt.show()
```

Practical 4 — Data Classification using Classification Algorithms

```
import pandas as pd
import matplotlib.pyplot as plt

from google.colab import files
uploaded = files.upload()

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

df = pd.read_csv("iris.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

X = df[["sepal_length", "sepal_width", "petal_length", "petal_width"]]
y = df["species"]

le = LabelEncoder()
y = le.fit_transform(y)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

model = DecisionTreeClassifier(random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("\nClassification Accuracy:")
print(f"{accuracy * 100:.2f}%")

cm = confusion_matrix(y_test, y_pred)
print("\nConfusion Matrix:")
print(cm)

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=le.classes_))

plt.figure(figsize=(8, 5))
plt.scatter(X_test["petal_length"], X_test["petal_width"], c=y_pred)
plt.xlabel("Petal Length")
plt.ylabel("Petal Width")
plt.title("Iris Flower Classification using Decision Tree")
plt.colorbar(label="Class")
plt.show()
```

Practical 5 — Data Clustering using Clustering Algorithms

```
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score
from scipy.cluster.hierarchy import dendrogram, linkage, fcluster

# Import directly from GitHub
url = "https://raw.githubusercontent.com/akmand/datasets/master/Mall_Customers.csv"
df = pd.read_csv(url)

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

X = df[["Annual Income (k$)", "Spending Score (1-100)"]]

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Elbow Method
inertia = []

for k in range(2, 11):
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(X_scaled)
    inertia.append(kmeans.inertia_)

plt.figure(figsize=(8, 5))
plt.plot(range(2, 11), inertia, marker="o")
plt.xlabel("Number of Clusters")
plt.ylabel("Inertia")
plt.title("Elbow Method")
plt.show()

# K-Means
kmeans = KMeans(n_clusters=5, random_state=42, n_init=10)
df["Cluster"] = kmeans.fit_predict(X_scaled)

silhouette = silhouette_score(X_scaled, df["Cluster"])
print("\nSilhouette Score:")
print(round(silhouette, 4))

print("\nClustered Data:")
print(df[["CustomerID", "Annual Income (k$)",
          "Spending Score (1-100)", "Cluster"]].head(10))

plt.figure(figsize=(8, 5))
plt.scatter(df["Annual Income (k$)"],
            df["Spending Score (1-100)"],
            c=df["Cluster"])
plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.title("Customer Segmentation using K-Means")
plt.colorbar(label="Cluster")
plt.show()

# Hierarchical Clustering
linked = linkage(X_scaled, method="ward")

plt.figure(figsize=(12, 6))
dendrogram(linked, truncate_mode="lastp", p=20)
plt.title("Hierarchical Clustering Dendrogram")
plt.xlabel("Customer Groups")
plt.ylabel("Distance")
plt.show()

hierarchical_clusters = fcluster(linked, t=5, criterion="maxclust")
df["Hierarchical_Cluster"] = hierarchical_clusters

print("\nHierarchical Clusters:")
print(df[["CustomerID", "Hierarchical_Cluster"]].head(10))

# If GitHub does not work, use:
# from google.colab import files
# uploaded = files.upload()
# df = pd.read_csv("Mall_Customers.csv")
```

Practical 6 — Linear Regression on Data Warehouse Data

```
import pandas as pd
import matplotlib.pyplot as plt

from google.colab import files
uploaded = files.upload()

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

df = pd.read_csv("store_sales_regression.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

X = df[["ad_spend_k", "footfall", "avg_discount_pct"]]
y = df["weekly_sales_k"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("\nIntercept:")
print(model.intercept_)

print("\nCoefficients:")
print(pd.Series(model.coef_, index=X.columns))

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("\nMean Squared Error:")
print(round(mse, 4))

print("\nR² Score:")
print(round(r2, 4))

plt.figure(figsize=(8, 5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Weekly Sales")
plt.ylabel("Predicted Weekly Sales")
plt.title("Actual vs Predicted Weekly Sales")
plt.show()
```

Practical 7 — Logistic Regression on Data Warehouse Data

```
import pandas as pd
import matplotlib.pyplot as plt

from google.colab import files
uploaded = files.upload()

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    accuracy_score, confusion_matrix, classification_report,
    roc_curve, roc_auc_score
)

df = pd.read_csv("customers_ml_logistic_regression.csv")

print("First 5 rows:")
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

X = df[[
    "age",
    "annual_income_k",
    "visits_per_month",
    "discount_used",
    "tenure_months"
]]

y = df["purchased"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

model = LogisticRegression(random_state=42)
model.fit(X_train_scaled, y_train)

y_pred = model.predict(X_test_scaled)
y_probability = model.predict_proba(X_test_scaled)[:, 1]

accuracy = accuracy_score(y_test, y_pred)

print("\nAccuracy:")
print(f"{accuracy * 100:.2f}%")

cm = confusion_matrix(y_test, y_pred)

print("\nConfusion Matrix:")
print(cm)

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

auc = roc_auc_score(y_test, y_probability)

print("\nROC-AUC Score:")
print(round(auc, 4))

fpr, tpr, thresholds = roc_curve(y_test, y_probability)

plt.figure(figsize=(8, 5))
plt.plot(fpr, tpr, label=f"ROC-AUC = {auc:.2f}")
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Logistic Regression")
plt.legend()
plt.show()
```

Practical 8 — Reading a CSV File and Data Analysis using Python Pandas

```
import pandas as pd
import matplotlib.pyplot as plt

from google.colab import files
uploaded = files.upload()

df = pd.read_csv("iris.csv")

print("First 5 rows:")
print(df.head())

print("\nLast 5 rows:")
print(df.tail())

print("\nDataset Shape:")
print(df.shape)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nStatistical Summary:")
print(df.describe())

print("\nSpecies Count:")
print(df["species"].value_counts())

print("\nAverage Measurements by Species:")

print(
    df.groupby("species")[
        ["sepal_length", "sepal_width",
         "petal_length", "petal_width"]
    ].mean()
)

df["species"].value_counts().plot(
    kind="bar",
    figsize=(8, 5)
)

plt.xlabel("Species")
plt.ylabel("Number of Samples")
plt.title("Number of Iris Samples by Species")
plt.xticks(rotation=0)
plt.show()
```